

How (not) to acquire exchange processes in Logical Phonology

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1 Introduction

- Since at least Meinhof (1912), phonologists have posited the existence of so-called *exchange*, *flip-flop*, or *polarity* alternations.

(1) Taxonomic definition of phonological exchange:

A phonological grammar has an *exchange* (with respect to some feature F) if, in some context $C __ D$ (where C, D are natural classes):

- $\begin{bmatrix} +F \\ \dots \end{bmatrix} \rightsquigarrow \{-F\}$, and
- $\begin{bmatrix} -F \\ \dots \end{bmatrix} \rightsquigarrow \{+F\}$.

- Exchanges can be thought of a special case of *circular chain shifts* (Moreton 2004).

(2) Dholuo (Tucker 1994):

	nom. sg.	pl.	
a.	gɔt	gɔdɛ	‘hill’
b.	kɛdɪ	kɛtɛ	‘twig’
c.	agɔkɔ	agɔgɛ	‘chest’
d.	hɪga	hike	‘year’

t ↔ d
k ↔ g

- This alternation applies to the other oral obstruents and in at least some loanwords (e.g., [kitæbu]–[kitepe] ‘book’).

(3) Itunyoso Triqui (DiCanio et al. 2020):

	inf.	3sg. prox.	
a.	ri ³²	rɪh ³	‘take out’
	a³ja ³²	a³jaɪh ³	‘read’
b.	a³toɪh ³	a³o ³	‘laugh’
	rãɪh ⁴	rã ³	‘buy’

V: ↔ ɪ

- Other hypothesized exchanges include:
 - Brussels Flemish Dutch (Zonneveld 1976): /u(:)/ ↔ /o(:)/
 - Czech (Wolfe 1970): /e:/ ↔ /i:/
 - English (Chomsky and Halle 1968:§4.4.3.8): “Rounding Adjustment”
 - Menomini (Bever 1967:86–94): /e/ ↔ /æ/
- However, in this talk we will argue:
 - Purely phonological exchanges may not exist.
 - Yet classical generative phonology can generate them, as can Logical Phonology.
 - If they do not in fact exist, there are various ways to explain their non-attestation.

2 Do phonological exchanges exist?

- Whether any of these languages actually have phonological exchanges is disputed:
 - Brussels Flemish Dutch (Moreton 2004)
 - Czech (Anderson and Browne 1973)
 - Dholuo (de Lacy 2012)
 - English (McCawley 1974)
 - Itunyoso Triqui (de Lacy 2020)
 - Menomini (Miner 1975)
- De Lacy (2012), for example, claims Dholuo does not involve a phonological exchange:
 - He notes that there are many non-alternating nouns (e.g., [cupæ]–[cupe] ‘bottle’).
 - He proposes new generalizations that do not require an exchange:
 - * The pl. devoices the last consonant in a polysyllabic, vowel-final root as in /kɛd-ε/ → [kɛtɛ] ‘twigs’.
 - * The nom.sg. devoices a word-final consonant as in /gɔd/ → [gɔt] ‘hills’.
 - If both ‘hill’ and ‘twig’ have an underlying /d/, there is no exchange to speak of.

- Thus evidence for phonological exchanges is murky at best:
 - Anderson and Browne (op. cit.) claim all synchronic chain shifts—not just exchanges—are actually morphophonological.
 - De Lacy (op. cit.) claims the existence of phonological exchanges has not been conclusively demonstrated.
- For sake of discussion—and in lieu of a compelling naturally-occurring example—we provide the following toy data set for “Fauxluo”, loosely inspired by Dholuo.

	sg.	du.	tr.	pl.	
(4)	a. nap	nap-ga	nab-lu	nap-a	‘cactus’
	b. nab	nab-ga	nap-lu	nab-a	‘igloo’

p ↔ b

3 Modeling phonological exchanges

3.1 Exchanges in classical generative phonology

- Chomsky and Halle (1968:355f.), who assume the existence of exchanges, note that modeling exchanges with ordinary ordered rules produces in an ordering paradox.

$$(5) \left[\begin{array}{c} +\text{LABIAL} \\ -\text{SONORANT} \\ +\text{VOICE} \end{array} \right] \longrightarrow \{-\text{VOICE}\} / __ [+LATERAL]$$

$$(6) \left[\begin{array}{c} +\text{LABIAL} \\ -\text{SONORANT} \\ -\text{VOICE} \end{array} \right] \longrightarrow \{+\text{VOICE}\} / __ [+LATERAL]$$

(7) Ordering paradox:

UR	nap-lu	nab-lu	UR	nap-lu	nab-lu
Rule (5):		naplu	Rule (6):	nablu	
Rule (6):	nablu	nablu	Rule (5):	naplu	naplu
SR	nablu	* nablu		* naplu	naplu

- Intuitively, what is needed is for (5–6) to apply simultaneously.
- C&H, following Bever (1967), accomplish this using α -notation.
- The following rule has the same effect as the simultaneous application of (5–6), generating the pattern in (4).

$$(8) \left[\begin{array}{c} +\text{LABIAL} \\ -\text{SONORANT} \\ \alpha\text{VOICE} \end{array} \right] \longrightarrow \{-\alpha\text{VOICE}\} / __ [+LATERAL]$$

3.2 Exchanges in Logical Phonology

- Logical Phonology (LP; e.g., Bale et al. 2014, 2020, Bale and Reiss 2018, Dabbous et al. 2021, 2025, Gorman and Reiss 2024, Reiss 2021) is an austere, formally rigorous theory of phonological computation.
- Features are assumed to be binary-valued (i.e., rather than privative).
- Segments are sets of valued features, written using curly brackets.
- Natural classes are similarly denoted, but are written using square brackets.
- Sets of valued features must be *consistent*.

(9) Consistency: If a segment is +F, it cannot also be –F, and vice versa.

- However featural underspecification is permitted—or more to the point, not forbidden—at any stage of the derivation.
- LP, following Poser (1982) and others, decomposes so-called feature-changing rules (e.g., voice assimilation) into two steps:¹

- In the first step, a *subtraction* rule (denoted by the $\setminus$ operator), essentially set difference, removes one or more valued features from a segment.

(10) Subtraction: $A \setminus B = \{cF \mid cF \in A \wedge cF \notin B\}$.

- In the second step, a *unification* rule (denoted by the $\sqcup$ operator), loosely based on similar operations in syntactic theory (e.g., Shieber 1986), adds a valued feature.

(11) Unification: $A \sqcup B = A \cup B$ if $A \cup B$ is consistent; otherwise, $A \sqcup B = A$.

Unification is similar to union, but applies vacuously—returning the left-side argument—when adding the valued feature would produce an inconsistent feature vector.

3.2.1 A failed attempt

- (8) is a feature-changing rule: it changes +VOICE to –VOICE and vice versa.
- Since LP does not have feature-changing *rules*, it does not have exchange *rules*.
- However, LP does have feature-changing *processes* implemented via a sequence of rules, usually one or more subtractions followed by one or more unifications.
- The following rule removes $\pm$ VOICE specifications from labial stops followed by a lateral.

$$(12) \left[\begin{array}{c} +\text{LABIAL} \\ -\text{SONORANT} \end{array} \right] \setminus \{ \alpha \text{VOICE} \} / \text{---} [+ \text{LATERAL}]$$

Once this rule applies, though, the underlying distinction between /p, b/ has been lost, and no subsequent rule can refer to it.

¹This brief review does not cover all rule types in LP; note also that subtraction and unification may be applied independently, as can be as seen in various examples below.

3.2.2 The Duke of York gambit

- Bever (1967:89f.) notes that exchange processes can also be generated via a *Duke of York* derivation (in the sense of Pullum 1976).
- Bever's construction uses three feature-changing ordered rules; an LP translation requires a sequence of three unifications and two subtractions. The basic idea is as follows:
 - Introduce an ad hoc feature specification at the start of the derivation to preserve a trace of the underlying VOICE distinction.
 - Use this ad hoc specification to condition the addition (i.e., unification) and subtraction of VOICE specifications.
 - Remove this ad hoc feature specification at the end of derivation.
- The first rule introduces the ad-hoc feature specification +X.²

$$(13) \left[\begin{array}{c} +\text{LABIAL} \\ -\text{SONORANT} \\ -\text{VOICE} \end{array} \right] \sqcup \{+X\} / \text{---} [+LATERAL]$$

- Then the following—repeated from (12)—removes $\pm\text{VOICE}$ specifications in this context.

$$(14) \left[\begin{array}{c} +\text{LABIAL} \\ -\text{SONORANT} \end{array} \right] \setminus \{\alpha\text{VOICE}\} / \text{---} [+LATERAL]$$

- The next two rules implement the exchange of VOICE:
 - The former is conditioned by the presence of the *ad hoc* +X.
 - The latter need not be: unification applies vacuously to targets specified for VOICE.

$$(15) [+X] \sqcup \{+\text{VOICE}\}$$

$$(16) [\] \sqcup \{-\text{VOICE}\}$$

- A final rule removes the *ad hoc* +X.

$$(17) [\] \setminus \{+X\}$$

²For simplicity, we assume that there are no $\pm X$ specifications either underlyingly or on the surface.

3.2.3 Opportunistic underspecification

- A slightly simpler approach to exchange processes in LP makes use of underspecification.
- In Fauxluo, there is a [p] $\leftrightarrow$ [b] exchange, but this does not imply that the segments involved are the fully-specified /p, b/.
- For concreteness, let us assume:
 - The final consonant in ‘cactus’ (sg. [nap]) is +LABIAL, –SONORANT, –CONTINUANT, –VOICE, and so on.
 - The final consonant in ‘igloo’ (sg. [nab]) is +LABIAL and –SONORANT but is not underlyingly specified for VOICE or CONTINUANT.
- The first rule adds a –VOICE specification to the root-final consonant of ‘igloo (tr.)’.

$$(18) \left[\begin{array}{c} +\text{LABIAL} \\ -\text{SONORANT} \end{array} \right] \sqcup \{-\text{VOICE}\} / \text{---} [+LATERAL]$$

This applies vacuously to ‘cactus (tr.)’, which is specified +VOICE.

- The second rule removes the –VOICE specification from the final consonant of ‘cactus (tr.)’.

$$(19) \left[\begin{array}{c} +\text{LABIAL} \\ -\text{SONORANT} \\ -\text{CONTINUANT} \end{array} \right] \setminus \{-\text{VOICE}\} / \text{---} [+LATERAL]$$

This does not affect ‘igloo (tr.)’, because its root-final consonant is not specified –CONTINUANT.

- The final two rules are context-free redundancy rules, and will apply vacuously when a earlier specification is present.

$$(20) [+LABIAL] \sqcup \{+VOICE\}$$

$$(21) [\] \sqcup \{-CONTINUANT\}$$

(22) Sample Fauxluo derivations (irrelevant specifications omitted):

UR	‘cactus (sg.)’ +L, –C, –V	‘cactus (tr.)’ +L, –C, –V	‘igloo (sg.)’ +L	‘igloo (tr.)’ +L
Rule (18):				+L, –V
Rule (19):		+L, –C		
Rule (20):		+L, –C, +V	+L, +V	
Rule (21):			+L, –C, +V	+L, –C, –V
SR	+L, –C, –V	+L, –C, +V	+L, –C, +V	+L, –C, –V

Summary thus far

- In classical generative phonology (§3.1):
 - α -notation, which is independently needed for dissimilation, generates exchange rules.
 - But exchange processes can also be modeled via the Duke of York gambit.
- In Logical Phonology:
 - There is no way to express exchange rules (§3.2.1).
 - But there are at least two ways to express exchange *processes* (§3.2.2–§3.2.3).
- This distinction between *rule* and *process* is not always made explicit:

‘Exchange’ has both descriptive/taxonomic and theoretical meanings, and the theoretical meanings are theory-specific. In its descriptive sense, the IT [Itunyoso Triqui] alternations can be called ‘exchanges’... However, the theoretical issue is, for any specific theory, whether that theory can generate the observations about IT reported in D [DiCanio et al. 2020] without employing ‘exchange mechanisms’. What exactly ‘exchange mechanisms’ are depends on the theory. (de Lacy 2020:29–30)

4 Filters and protocols

- If phonological exchanges exist, there are a plethora of devices to generate them.
- But if phonological exchanges are unattested—and perhaps *unattestable*—does this mean our models are too powerful?

4.1 The diachronic filter

- For a phonological pattern to be attested, it necessarily must:
 - be formalizable within whatever model of phonological Universal Grammar (UG) one assumes, and
 - make it through the *diachronic filter*:
 - * It must follow from plausible series of sound changes and/or re-analyses,
 - * beginning from plausible initial conditions.
- It seems plausible this filter is responsible for the non-attestation of phonological exchanges, given the abstractness, complexity, and phonetic unnaturalness of the LP analyses, and if so there is no need to complexify UG with this apparent typological gap (e.g., Kaplan 1995).

4.2 LAD protocols

- Without discounting the diachronic filter account, there is another possibility: perhaps the pattern is unacquirable.
- Kiparsky (1968, et. seq.) and Kenstowicz and Kisseberth (1977: ch. 1), among others, make proposals to limit the apparent abstractness of phonological derivations.

(23) ALTERNATION CONDITION (Kiparsky 1973:65): Neutralization processes cannot apply to all occurrences of a morpheme.

- It is not at all clear to us what the ontological status of (23)—and similar proposals made during the post-*SPE* abstractness crisis—is for K and K&K and contemporaries:
 - Is it a negative heuristic for the analyst?
 - Is it a constraint on what grammars can be computed?
 - Is it a defeasible inductive bias encoded within the LAD?
- In lieu of a complete theory of the phonological LAD, we propose explicit *LAD protocols* that govern analyses proffered by the LAD. These are intended as a contribution to the theory of phonological *acquisition*, not phonological *computation*.
- Recall that the Duke of York analysis (§3.2.2) uses an ad hoc feature to track segments’ the underlying VOICE specifications. If the LAD is unable to posit ad hoc features that never surface, then this type of DoY derivation will be unattestable.

(24) NO WANDERING TARGETS: Any valued feature added to (the correspondent of) a segment by a unification rule must be a member of the generalized union of all valued features found in all allophones of that segment.

- (24) does not rule out all “false steps” (Zwicky 1974) or DoY derivations, just those that implicate a valued feature without any surface correspondents.

5 Conclusions

- There is of yet no conclusive evidence for the existence of phonological exchanges.
- Phonological exchange *rules* can be generated by the α -notation of classical generative phonology, whereas LP can generate phonological exchange *processes* in various ways.
- We propose that their apparent non-existence may reflect:
 - the diachronic filter, or
 - properties of the phonological LAD.

- Exchanges appear to be *computable* in LP, but their non-existence suggests they may be *unattestable* either because of the diachronic filter or because they are *unacquirable*.

(25) The cone of attestation (after Hale and Reiss 2008:4):

Attested $\subset$ Attestable $\subset$ Acquirable $\subset$ Computable $\subset$ Statable

- We believe these proposals are an improvement on the vague intuitions about elegance or plausibility commonly found in this literature (cf. Zwicky 1974, Pullum 1976).
- These proposals clarify epistemological and ontological issues—those faced by the linguist and those faced by the child—confounded in earlier critiques of phonological abstractness.
- The situation is somewhat different in Optimality Theory, where *anti-faithfulness* constraints (Alderete 2001) are used to model exchanges.

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